

MechMavrix CivicRoute

Evidence-driven civic complaint routing

Citizens report the problem. CivicRoute determines where it goes, how urgent it is, and what happens next.

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Primary subproblem: Routing

ROUTING WITHOUT GUESSWORK

- 1 Photo + GPS
- 2 AI-assisted classification
- 3 Server-side context
- 4 Priority + jurisdiction
- 5 Tracking + follow-through

Problem Statement

The problem is not reporting — it is routing.

1

Who owns this location?

Changing ward / authority boundaries create confusion.

2

Which department should act?

Citizens often guess the department and misroute complaints.

3

How urgent is it?

Votes alone do not capture public-risk context.

4

What happens after submission?

Poor follow-through reduces visibility and trust.

Citizens identify the issue; CivicRoute determines ownership, urgency and follow-through.

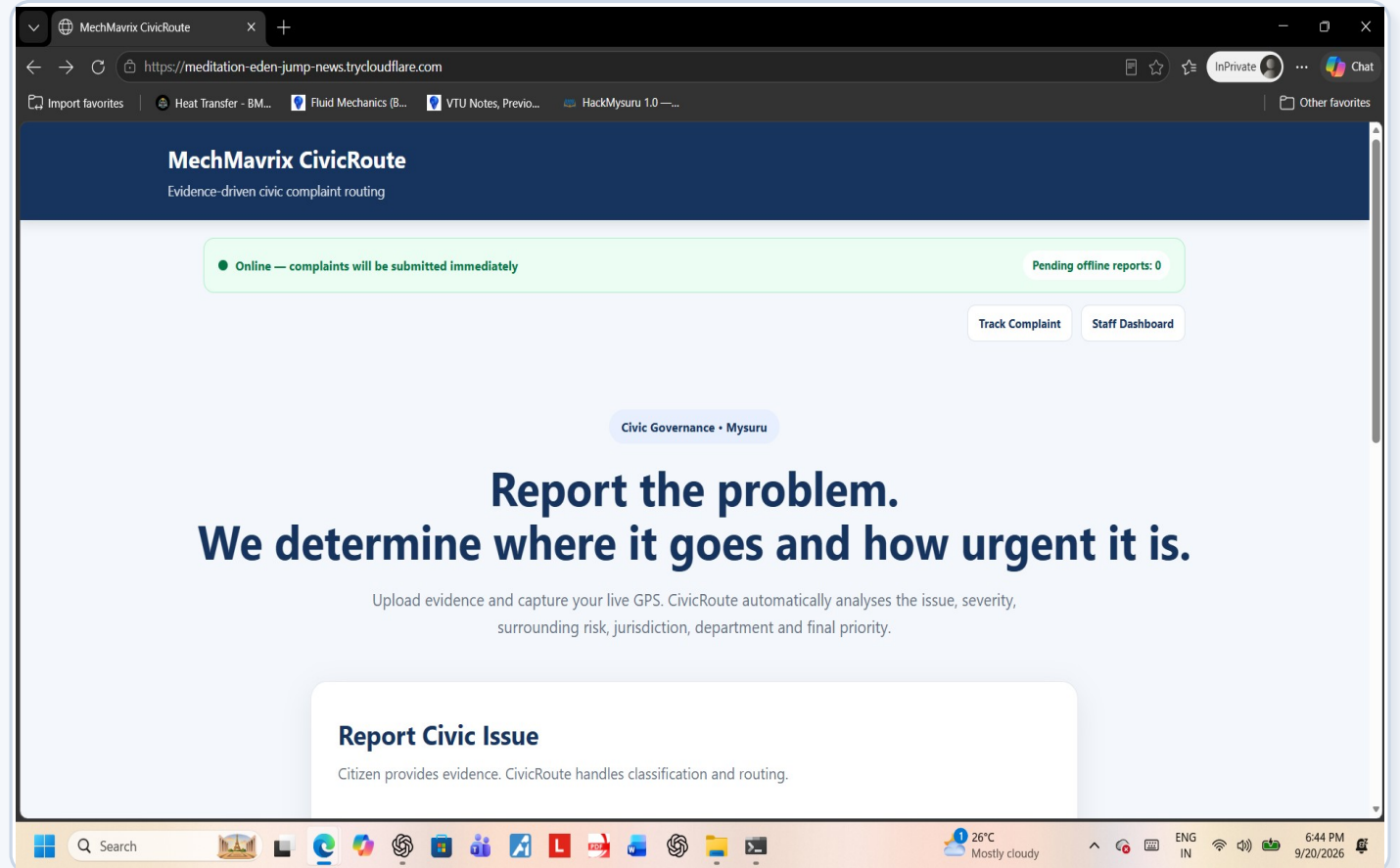
CivicRoute: minimal citizen input, maximum routing context

The citizen gives only what they actually know.

CITIZEN INPUT

- 1 Complaint photo
- 2 Current GPS
- 3 Optional description

Citizen does NOT choose authority, department, severity or priority.



SYSTEM DECIDES: ISSUE • SEVERITY • RISK • JURISDICTION • DEPARTMENT • PRIORITY

System architecture

Important decisions are recomputed on the backend - not trusted from the browser.



Trust boundary

- Frontend address / risk / priority values are not authoritative inputs.
- Backend independently derives location context and routing decisions.
- Failure to verify enough context becomes REVIEW REQUIRED instead of silent guessing.

EXPLAINABLE • VERSIONED • REVIEWABLE

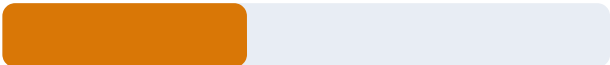
Decision logic: transparent priority + safe fallbacks

Priority is deterministic; uncertain AI or routing context is explicitly surfaced.

100-point priority model

Location risk  50 pts

Problem type  30 pts

Severity  20 pts

Why location gets 50%

The same issue can have different public impact near a hospital, school, traffic signal, bus station or marketplace.

Safety gates

- AI ambiguity** Low confidence / small margin → REVIEW REQUIRED
- Jurisdiction uncertainty** No active trusted rule → Jurisdiction Review Queue
- Duplicate candidate** Nearby recent match → review, not automatic spam label
- Versioned rules** Effective dates + provenance metadata before activation

AI assists the workflow - it does not own the truth

Local CLIP zero-shot inference helps classify visible civic issues and severity.

AI layer

OpenAI CLIP ViT-B/32

- Zero-shot issue classification
- Visual severity estimation
- Confidence + margin checks
- Human review fallback

No production accuracy percentage is claimed.

What the system deliberately does NOT claim

- No real-time crowd-density detection
- No legal proof of municipal GIS boundaries
- No forensic proof that a photo was captured at the submitted GPS
- No automatic accusation of spam from a nearby report
- No existing integration with official municipal production systems

AUTOMATE WHEN EVIDENCE IS SUFFICIENT • REVIEW WHEN IT IS NOT

Offline resilience: preserve first, synchronize later

Weak connectivity should not cause a citizen report to disappear.



Implemented with

Service Worker app-shell cache

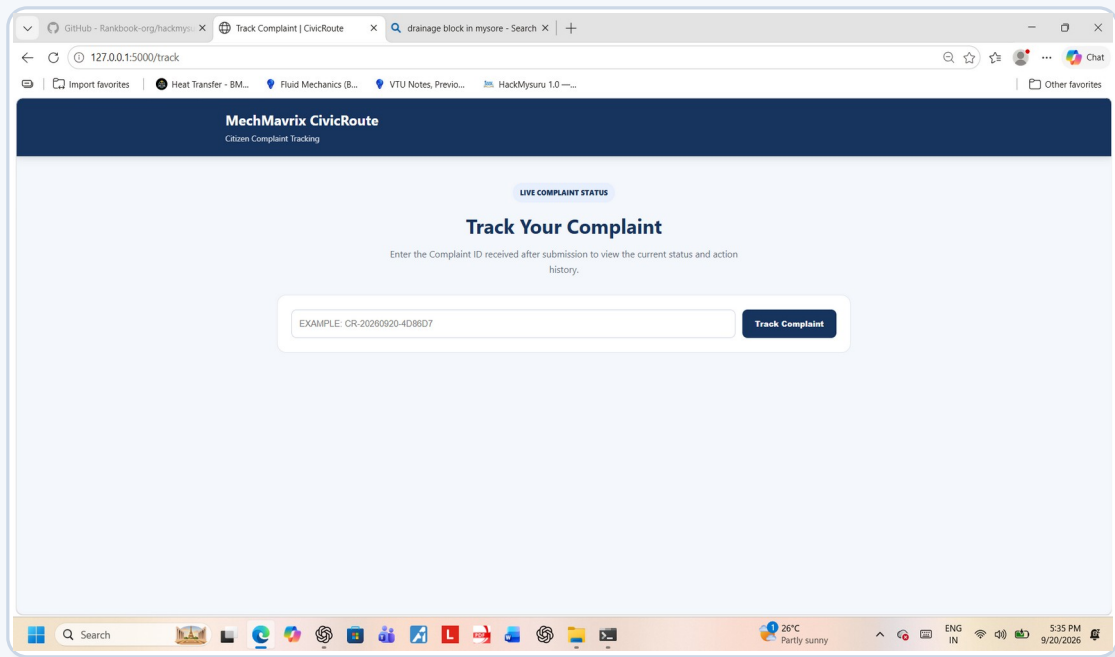
IndexedDB pending reports

Auto-sync on online event

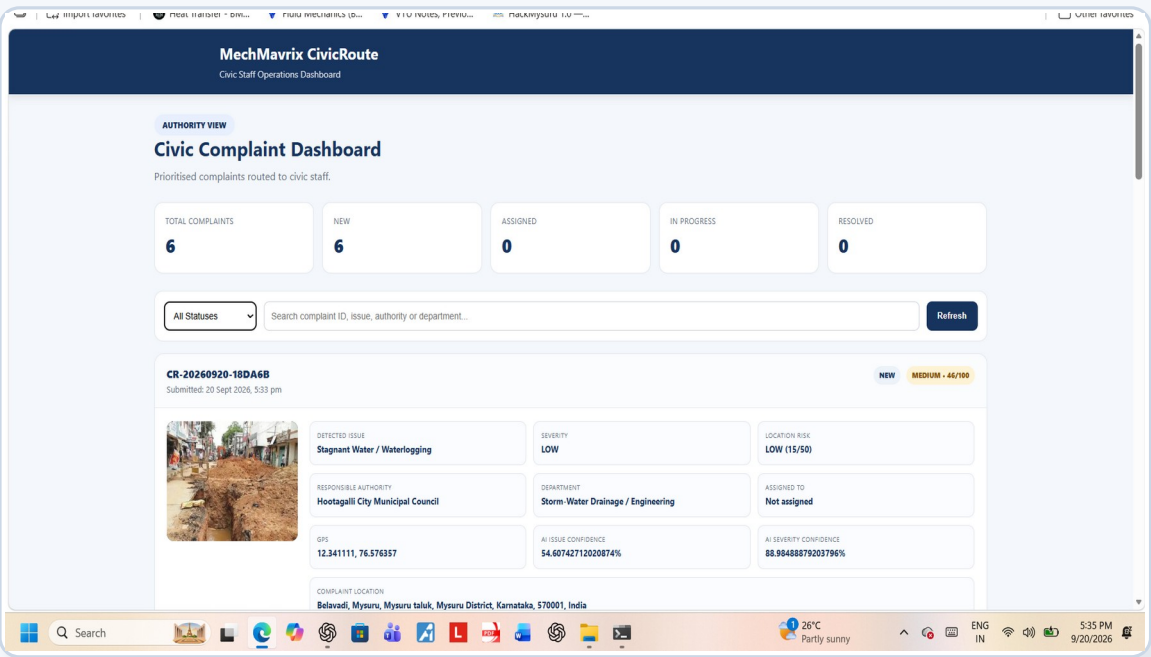
Offline limitations: fresh map tiles, reverse geocoding, AI inference and final routing wait for reconnection.

Follow-through: tracking for citizens, workflow for staff

Routing is useful only if the complaint remains visible after submission.



CITIZEN: STATUS • AUTHORITY • HISTORY



STAFF: FILTER • ASSIGN • RESOLVE

Complaint lifecycle



Validation and known limitations

We tested the core decision logic and documented what the MVP cannot guarantee.

17/17 Core routing & bad-input checks

12/12 Boundary provenance checks

6/6 Boundary versioning checks

PASS Offline reload + queue + reconnect sync

Current MVP limitations

- Locality rules are demonstration rules, not legal GIS polygons.
- OpenStreetMap / Nominatim / Overpass coverage can be incomplete or rate-limited.
- CLIP zero-shot scores are not calibrated civic-risk probabilities.
- Staff dashboard has no production authentication / RBAC yet.
- SQLite + local uploads are MVP choices, not production scaling architecture.

UNCERTAINTY IS EXPOSED, NOT HIDDEN

CivicRoute is live, public, documented and demo-ready

Citizens report the evidence. CivicRoute handles routing, priority, duplicate protection, tracking and follow-through.

1

Live MVP

<https://meditation-eden-jump-news.trycloudflare.com>

2

Public GitHub

github.com/Abhishek-Nayak09/MechMavrix-CivicRoute-HackMysuru

3

Demo video

08:41 continuous walkthrough • Google Drive

4

Decision log

1-page decisions + trade-offs in docs/decision-log.md

Thank you

TEAM MECHMAVRIX